AI Co-Scientist for Particle Physics

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Overview

1- About Project
2- Terminologist Agent
3- Table Agent
4- Graph Agent

What is the ....

problem?

Building a multi-agent generative AI system to help physicist search for physics beyond the Standard Model

purpose?

- Understand particle physics papers
- Focus on studies from the ATLAS and CMS experiments
- Read tables and graphs inside these papers
- Be able to answer questions to textual content, tables and figures.
- Compare results between different studies in same experiments and then between ATLAS and CMS
- And even calculate error margins

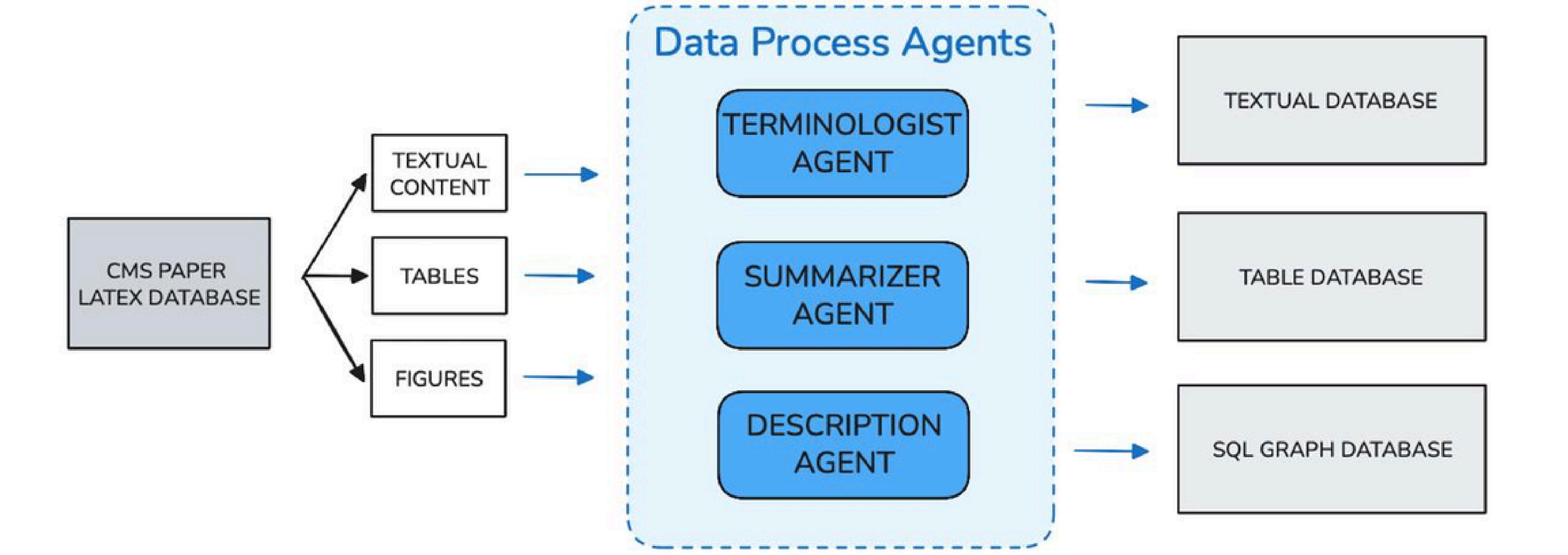
Methodology

Input:
CMS papers (PDF / LaTeX)

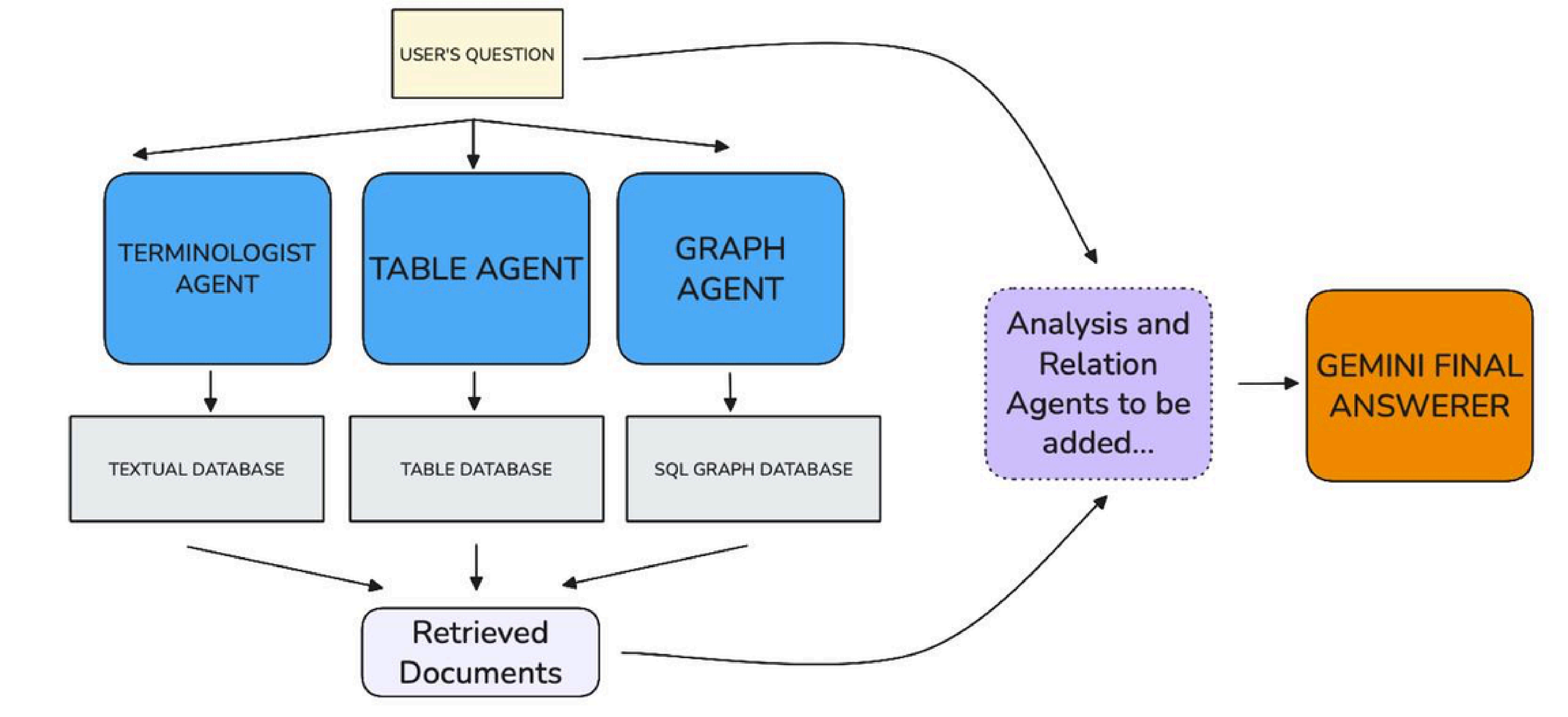
Processing:
Extracted textual content, tables, and figures are processed using Data Processing Agents and stored in separate databases.

Retrieval:
Relevant documents are retrieved using the Terminologist, Table, and Graph agents, then passed to further agents to generate the final answer.

General Database Generation Process



General Retrieval and Q/A Process



Results / Progress

EUROPEAN ORGANIZATION FOR NUCLEAR RESEARCH (CERN)

CERN-EP-2024-206
2025/02/20

CMS-SUS-23-008

Search for dark matter produced in association with a pair of bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

The CMS Collaboration*

Feb 2025

Question:
How does the design of the CMS detector, specifically the combination of the ECAL, HCAL, and muon systems inside the solenoid, contribute to the accurate reconstruction of missing transverse momentum?

Okay, I'll answer your question about how the CMS detector's design contributes to the accurate reconstruction of missing transverse momentum, using the provided context excerpts.

Based on excerpt [1]:

CMS Detector Overview: "The central feature of the CMS apparatus is a superconducting solenoid of 6m internal diamete r, providing a magnetic field of 3.8T. Within the solenoid volume are a silicon pixel and strip tracker, a lead tungstate crysta l electromagnetic calorimeter (ECAL), and a brass and scintillator hadron calorimeter (HCAL), each composed of a barrel and two endcap sections. Forward calorimeters, made of steel and quartz fibers, extend the pseudorapidity (pseudorapidity) coverage provided by the barrel and endcap detectors. Muons are detected in gas-ionization chambers embedded in the steel flux-return yoke outside the solenoid."

Based on excerpt [2]:

Missing Transverse Momentum Calculation: "As mentioned in Section <ref>, the negative vectorial transverse momentum sum of all PF candidates is used to calculate transverse momentumvecmiss with all jet energy corrections are applied."

Combining this information, here's how the CMS detector's components contribute to missing transverse momentum reconstruction:

1. **Solenoid and Magnetic Field:** The 3.8T magnetic field provided by the solenoid is crucial for measuring the momentum o f charged particles. The curvature of a charged particle's track in the magnetic field is directly related to its momentum. Acc urate momentum measurements are essential for calculating the vectorial sum of transverse momentum.

2. **Tracker:** Although not directly mentioned in relation to ptmiss, the silicon tracker is key for precisely measuring the trajectories of charged particles, allowing for the proper identification and measurement of charged particle momenta. This w ill improve the accuracy of ptmiss calculations.

3. **ECAL (Electromagnetic Calorimeter):** The ECAL measures the energy of electrons and photons. Accurate energy measurements of these electromagnetically interacting particles are essential for the overall energy balance in the transverse plane, which is used to infer missing transverse momentum. The ECAL is made of lead tungstate crystals.

4. **HCAL (Hadron Calorimeter):** The HCAL measures the energy of hadrons (particles made of quarks, such as protons, neutrons, and pions). The HCAL, made of brass and scintillator, is crucial for capturing the energy of jets, which are sprays of hadron s produced in high-energy collisions. Correcting jet energy is explicitly mentioned in excerpt [2] as being important for missing transverse momentum calculation.

5. **Muon System:** Muons are weakly interacting particles that can pass through the calorimeters. The muon system, located outside the solenoid, identifies and measures the momentum of muons. Because muons often escape the calorimeters, accurately measuring their momentum is vital for the overall momentum balance.

6. **Particle Flow Algorithm (PF):** Although not explicitly defined in the provided excerpts, the reference to "all PF cand idates" [2] implies that the CMS experiment uses a particle flow algorithm. This algorithm combines information from all subdetecto rs (tracker, ECAL, HCAL, muon system) to reconstruct all individual particles in the event (charged hadrons, photons, electrons, muons, ne utral hadrons). The missing transverse momentum is then calculated as the negative vector sum of the transverse moment a of all reconstructed particles.

In summary, the CMS detector's design, with its central magnetic field, high-resolution tracker, precise electromagnetic and hadronic calorimeters, and dedicated muon system, enables a comprehensive reconstruction of all detectable particles produced in a collision. By accurately measuring the momentum and energy of these particles, the missing transverse momentum, which is an indicator of non-interacting particles like dark matter, can be reliably inferred.

ChatGPT 4o

2408.17336v2.pdf
PDF

How does the design of the CMS detector, specifically the combination of the ECAL, HCAL, and muon systems inside the solenoid, contribute to the accurate reconstruction of missing transverse momentum?

The design of the CMS detector—with the electromagnetic calorimeter (ECAL), hadron calorimeter (HCAL), and muon systems all located inside the solenoid—allows for precise measurement of all visible particles' momenta, which is critical for accurately reconstructing missing transverse momentum (p_t^miss) by ensuring minimal energy escapes undetected from within the magnetic field volume.

Sources

1. Designed the architecture and agent workflow
2. Developed and tested the Terminologist Agent, successfully converting LaTeX symbols and equations into interpretable natural language explanations
3. Table and Graph agent development completed.
4. Improve multi-agent coordination and performance validation
5. Deploy system on CMS & ATLAS analysis databases

Data Sources

CMS Papers

In this study, Supersymmetry papers published by the CMS Collaboration are used. Their LaTeX source codes are stored in a database.

CMS-SUS LaTeX Files

CMS-SUS-10-002.tex	CMS-SUS-11-020.tex	CMS-SUS-12-028.tex	CMS-SUS-14-004.tex	CMS-SUS-15-004.tex
CMS-SUS-10-003.tex	CMS-SUS-11-021.tex	CMS-SUS-13-002-003.tex	CMS-SUS-14-005.tex	CMS-SUS-15-005.tex
CMS-SUS-10-004.tex	CMS-SUS-11-022.tex	CMS-SUS-13-003.tex	CMS-SUS-14-006.tex	CMS-SUS-15-006.tex
CMS-SUS-10-005.tex	CMS-SUS-11-024.tex	CMS-SUS-13-004.tex	CMS-SUS-14-007.tex	CMS-SUS-15-007.tex
CMS-SUS-10-006.tex	CMS-SUS-11-028.tex	CMS-SUS-13-006.tex	CMS-SUS-14-009.tex	CMS-SUS-15-008.tex
CMS-SUS-10-007.tex	CMS-SUS-12-001.tex	CMS-SUS-13-007.tex	CMS-SUS-14-010.tex	CMS-SUS-15-009.tex
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CMS-SUS-11-003.tex	CMS-SUS-12-006.tex	CMS-SUS-13-019.tex	CMS-SUS-14-019.tex	CMS-SUS-16-008.tex
CMS-SUS-11-010.tex	CMS-SUS-12-010.tex	CMS-SUS-13-023.tex	CMS-SUS-14-020.tex	CMS-SUS-16-009.tex
CMS-SUS-11-011.tex	CMS-SUS-12-011.tex	CMS-SUS-13-024.tex	CMS-SUS-14-021.tex	CMS-SUS-16-017.tex
CMS-SUS-11-013.tex	CMS-SUS-12-014.tex	CMS-SUS-14-001.tex	CMS-SUS-14-022.tex	CMS-SUS-16-032.tex
CMS-SUS-11-016.tex	CMS-SUS-12-017.tex	CMS-SUS-14-002.tex	CMS-SUS-15-002.tex	CMS-SUS-16-033.tex
CMS-SUS-11-018.tex	CMS-SUS-12-024.tex	CMS-SUS-14-003.tex	CMS-SUS-15-003.tex	CMS-SUS-16-034.tex

CMS-SUS-16-035.tex	CMS-SUS-16-051.tex	CMS-SUS-18-005.tex	CMS-SUS-20-001.tex
CMS-SUS-16-036.tex	CMS-SUS-17-001.tex	CMS-SUS-18-006.tex	CMS-SUS-20-002.tex
CMS-SUS-16-037.tex	CMS-SUS-17-003.tex	CMS-SUS-18-007.tex	CMS-SUS-20-003.tex
CMS-SUS-16-038.tex	CMS-SUS-17-004.tex	CMS-SUS-19-001.tex	CMS-SUS-20-004.tex
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CMS-SUS-16-041.tex	CMS-SUS-17-007.tex	CMS-SUS-19-004.tex	CMS-SUS-21-003.tex
CMS-SUS-16-042.tex	CMS-SUS-17-008.tex	CMS-SUS-19-005.tex	CMS-SUS-21-004.tex
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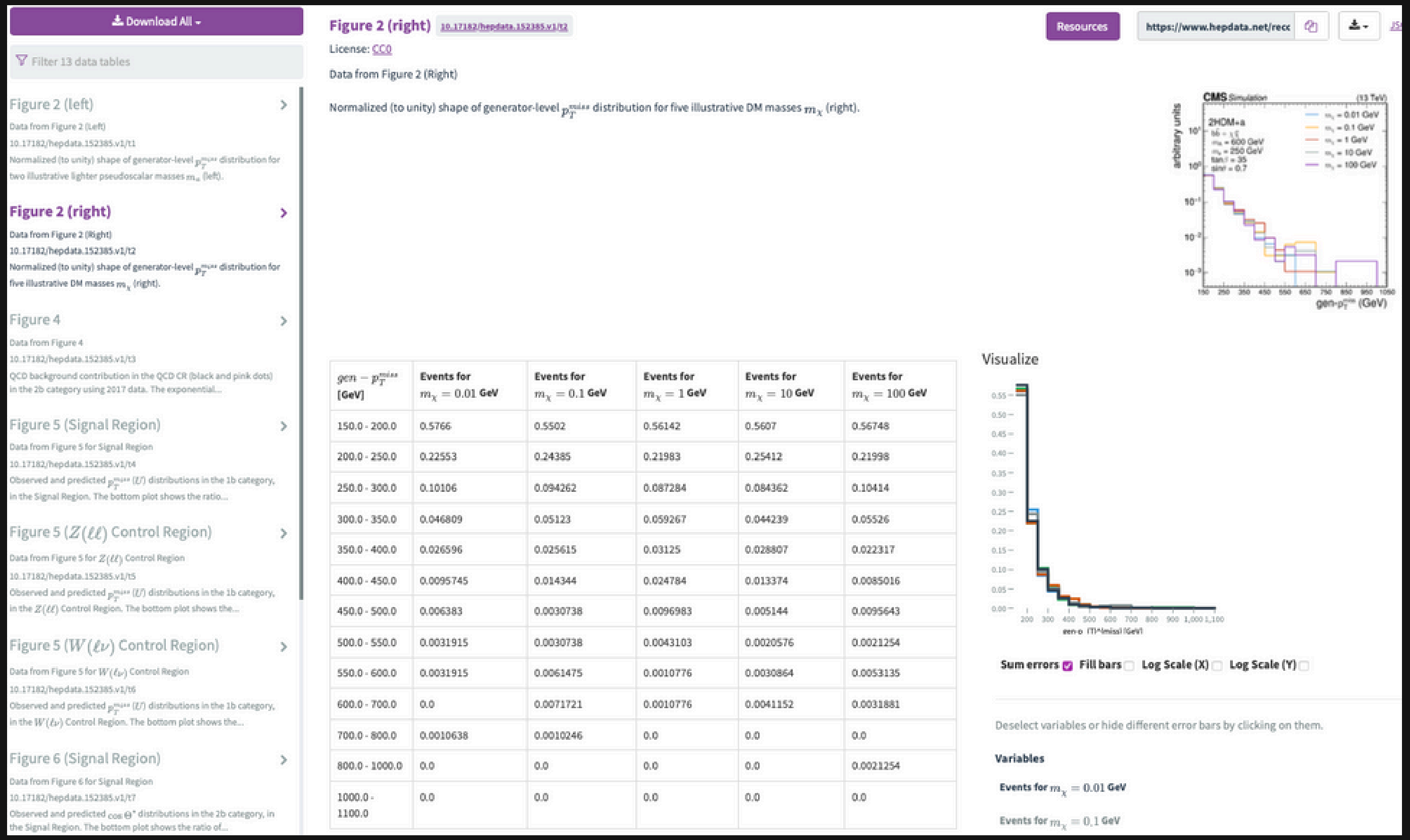
CMS-SUS-16-048.tex	CMS-SUS-18-002.tex	CMS-SUS-19-011.tex	CMS-SUS-23-015.tex
CMS-SUS-16-049.tex	CMS-SUS-18-003.tex	CMS-SUS-19-012.tex	CMS-SUS-24-001.tex
CMS-SUS-16-050.tex	CMS-SUS-18-004.tex	CMS-SUS-19-013.tex	

Figures from Papers

Numerical data and metadata of the figures are obtained from HEPData. Figure data is available for 21.6% of the papers in HEPData. This data is essential for building a system capable of answering questions based on the graphs in the figures.

HepData Website



TERMINOLOGIST AGENT

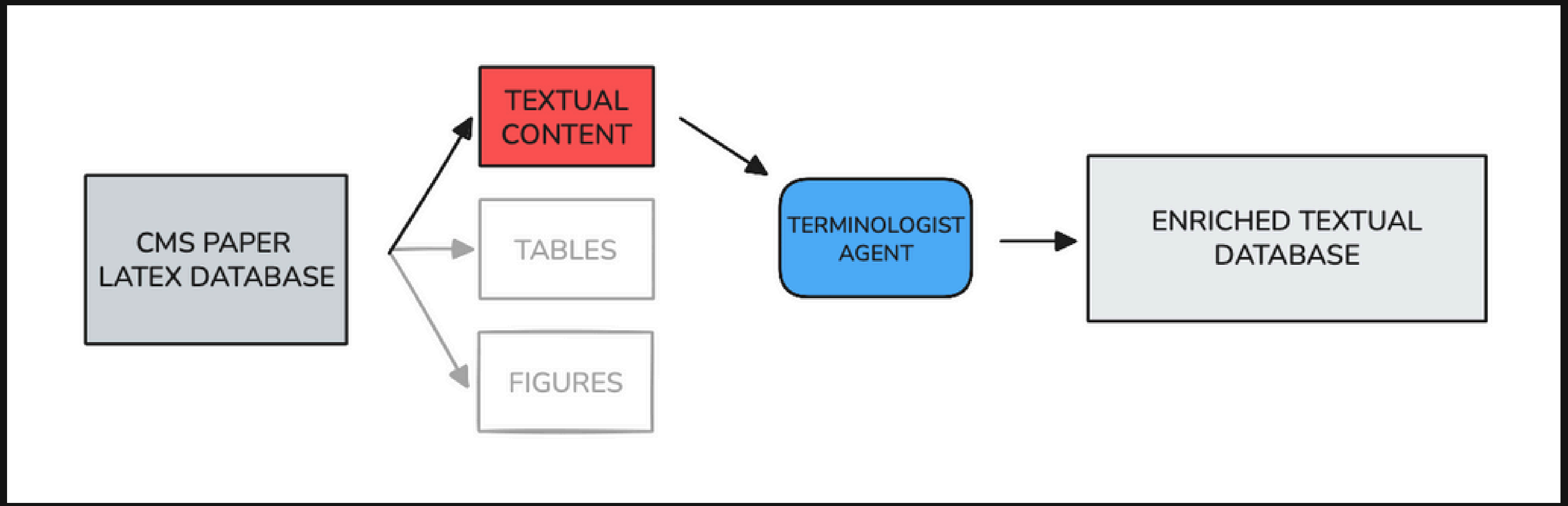
Terminologist Data Preparation and the Terminologist Agent

Agent Definition: This agent maps the LaTeX format of physics terms and equations to their natural language counterparts. For example: \pT → Transverse Momentum. It then generates both a brief explanation and a contextual use case for each term, based on the section of the paper where it appears. Additionally, it interprets symbolic expressions and mathematical equations, allowing other agents to understand and reason about their meaning and role in the analysis.

LaTeX to Natural Language

```
Enriched: \bbbar+\ptmiss ➡bottom-antibottom plus missing transverse momentum
Enriched: \Pp\Pp ➡proton-proton
Enriched: \PQt/\PAQt ➡top quark / anti-top quark
Enriched: \ttbar ➡top quark pair
Enriched: \bbbar ➡bottom-antibottom quark pair
Enriched: \ptvecmiss ➡missing transverse momentum vector
Enriched: \ptmiss ➡missing transverse momentum magnitude
Enriched: \GeV ➡Giga electron volt
Enriched: \TeV ➡Teraelectronvolt
Enriched: \PX ➡X particle
Enriched: \fbinv ➡inverse femtobarn
Enriched: \Pb ➡b-jet
Enriched: \mh = 125\GeV ➡mass of the Higgs boson equals 125 GeV
Enriched: \PGc ➡Dirac fermion
```

Textual Database Preperation Process



Purpose of Terminologist Agent

Problem:

Scientific physics papers are full of:

- Complex math symbols
- LaTeX-style notations (e.g., \ptmiss, \mHpm, p_T, \bbbar)
- Abbreviations and special formatting

⚠️ These terms make sense to humans but to a computer, they're just meaningless character strings.

Solution: Terminologist Agent

A specialized AI agent that:

1. Finds all special terms in the paper
2. Uses the surrounding context (section text)
3. Asks an LLM (e.g., Gemini):
4. "What does this term mean and how is it used?"
5. Generates a structured explanation in JSON format

Result:

- This process gives meaning to LaTeX symbols.
- Now, all other AI agents—like table, graph agents, or QA bots—can understand and interpret the same terms consistently.
- They speak a "shared language," enabling deeper and more accurate analysis of physics papers.

An Example From Terminologist Data


```
{
  "id": "cms-sus-23-008_S0_B5",
  "term": "\\fbinv",
  "start": 2060,
  "end": 2066,
  "explanation": "An inverse femtobarn is a unit of measurement for integrated luminosity. Integrated luminosity is a
  measure of the total amount of data collected in a particle physics experiment. Specifically, it represents the integral
  of the instantaneous luminosity over time. Luminosity, in turn, is a measure of the rate of collisions in a particle collider.
  Therefore, a higher integrated luminosity means more collisions, and thus more data, which increases the chances of observing
  rare processes. One inverse femtobarn corresponds to the amount of data needed to produce, on average, one event for a process
  with a cross-section of one femtobarn.",
  "Usecase": "In this context, the value of 138 inverse femtobarns represents the total amount of proton-proton collision data
  collected by the CMS detector between 2016 and 2018. This large amount of data is used to search for dark matter particles produced in
  association with bottom quarks. A larger integrated luminosity allows for a more sensitive search for rare events.",
  "pronunciation": "inverse femto barn",
  "natural_form": "inverse femtobarn"
},
```


Terminologist provides
REASONING!

Performance Test of Terminologist Agent

Question → How does the design of the CMS detector, specifically the combination of the ECAL, HCAL, and muon systems inside the solenoid, contribute to the accurate reconstruction of missing transverse momentum?

FINAL ANSWER (step-by-step, structured reasoning):

ANSWER:
Okay, let's break down how the CMS detector's design contributes to accurate missing transverse momentum (MET) reconstruction. This is a crucial aspect of many analyses, especially those searching for new physics involving weakly interacting particles like dark matter candidates.

Step 1: Relevant Terms and Why

The question specifically asks about the ECAL, HCAL, muon systems, and the solenoid magnet. While the provided text focuses on the 2HDM+a model parameters and dark matter searches, it doesn't directly define or explain the sub-detector components of CMS. To properly address this question, I will rely on my knowledge of the CMS detector design. This will allow me to describe how these components collectively contribute to MET reconstruction. I will also include the definition of MET.

* *ECAL*
  * Natural Form: Electromagnetic Calorimeter
  * Context: (From outside Knowledge) The ECAL is designed to precisely measure the energy of electrons and photons through electromagnetic showers in a dense crystal material (typically lead tungstate).
  * Detailed Explanation: When electrons or photons interact with the ECAL material, they initiate a cascade of secondary particles (an electromagnetic shower). The ECAL measures the energy deposited by these showers. Its high granularity (finely segmented structure) allows for precise position measurements of the particles.
  * Usecase: Crucial for identifying and measuring the energy of electrons and photons produced in collisions. This information is directly used in calculating the total transverse momentum of all visible particles.

* *HCAL*
  * Natural Form: Hadronic Calorimeter
  * Context: (From outside Knowledge) The HCAL measures the energy of hadrons (particles made of quarks and gluons, such as protons, neutrons, pions, and kaons). It is located outside the ECAL.
  * Detailed Explanation: Hadrons interact strongly and produce hadronic showers in the HCAL. The HCAL is typically made of alternating layers of dense absorber material (e.g., steel or brass) and active detector material (e.g., scintillator or gas chambers). The HCAL is less granular than the ECAL.
  * Usecase: Essential for measuring the energy of jets (sprays of hadrons) originating from quarks and gluons. Accurately measuring jet energies is vital for reconstructing the total transverse momentum.

* *Muon System*
  * Natural Form: Muon Detector System
  * Context: (From outside Knowledge) Muons are heavier cousins of electrons that can penetrate through the calorimeters. The muon system, located outside the calorimeters, is designed to identify and measure the momentum of muons.
  * Detailed Explanation: The muon system typically consists of multiple layers of detectors (e.g., drift tubes, cathode strip chambers, resistive plate chambers) that track the passage of muons. The bending of muon tracks in the magnetic field allows for momentum measurement.
  * Usecase: Since muons are penetrating particles, their momentum must be measured independently of the calorimeters. Accurate muon momentum measurement is essential for MET calculation.

* *Solenoid*
  * Natural Form: Solenoid Magnet
  * Context: (From outside Knowledge) A strong magnetic field is crucial for measuring the momentum of charged particles. The CMS solenoid provides a strong axial magnetic field (3.8 T).
  * Detailed Explanation: The solenoid is a large cylindrical magnet that surrounds the inner detectors (tracker, ECAL, and HCAL). The magnetic field causes charged particles to curve as they travel through the detector. The radius of curvature is proportional to the particle's momentum.
  * Usecase: The magnetic field is essential for measuring the momentum of charged particles. Without it, you can only measure the energy deposited in the calorimeters, but not the individual particle momenta.

* *Missing Transverse Momentum (MET)*
  * Natural Form: Missing Transverse Momentum
  * Context: (From outside Knowledge) In proton-proton collisions, the total transverse momentum should be zero, as the colliding protons have negligible transverse momentum. However, particles that do not interact with the detector, such as neutrinos or hypothetical dark matter particles, can carry away momentum undetected. This leads to an apparent imbalance in the measured transverse momentum.
  * Detailed Explanation: MET is a vector quantity calculated as the negative vector sum of the transverse momenta of all detected particles in an event. Mathematically: MET = - (sum of pT_i) where pT_i is the transverse momentum vector of the i-th detected particle. The magnitude of the MET vector is often referred to as Missing Transverse Energy (MET or ETmiss).
  * Usecase: MET is a key signature of particles that escape detection. It is widely used in searches for neutrinos, dark matter, and other weakly interacting particles. It is also used in precision measurements of Standard Model processes.

Continiues -->

Performance Test of Terminologist Agent

Question → How does the design of the CMS detector, specifically the combination of the ECAL, HCAL, and muon systems inside the solenoid, contribute to the accurate reconstruction of missing transverse momentum?

Step 2: Answering the Question

The question asks how the ECAL, HCAL, and muon systems, within the solenoid's magnetic field, contribute to accurate MET reconstruction. The key here is that MET is inferred; it's not directly measured. It's calculated from what is measured.

Step 3: Step-by-Step Explanation

Let's build this up piece by piece:

1. *The Goal: Measuring Everything You Can.* The foundation of accurate MET reconstruction is measuring the transverse momentum of all detectable particles produced in the collision as precisely as possible. Any error in measuring a single particle's momentum directly propagates into the MET calculation.

2. *Charged Particle Momentum Measurement (Tracker + Solenoid):* The tracker, located closest to the interaction point inside the solenoid, precisely measures the trajectories of charged particles. The solenoid provides the crucial magnetic field that bends the paths of these particles. The curvature of the track is directly related to the particle's momentum. This provides the initial measurement of charged particles.

3. *Electromagnetic Energy Measurement (ECAL):* Electrons and photons deposit almost all of their energy in the ECAL. The ECAL's high granularity allows for precise measurement of their energy and position. This energy measurement is then combined with the track information from the tracker (when applicable, for charged particles) to provide a more complete picture of the particle's transverse momentum. For photons, which leave no track, the ECAL energy and position provides the entire transverse momentum measurement.

4. *Hadronic Energy Measurement (HCAL):* Hadrons (primarily those within jets) deposit energy in the HCAL. The HCAL provides a measurement of the energy of these particles. The HCAL's energy measurement is combined with tracker information (for charged hadrons) to reconstruct the momentum of the hadrons and, subsequently, the jets they form. Jet reconstruction algorithms rely heavily on both calorimeter and tracking information to accurately determine the jet's momentum.

5. *Penetrating Muon Momentum Measurement (Muon System):* Muons pass through the calorimeters without depositing all their energy. The muon system tracks these particles and, using the magnetic field (either from the solenoid leakage field or its own magnetic field, depending on the detector design), measures their momentum independently.

6. *Putting it All Together (MET Calculation):* The measured transverse momenta of all reconstructed objects (electrons, photons, muons, charged hadrons, and neutral hadrons) are vectorially summed. The negative of this sum is the MET vector.

7. *Why This Design Matters for Accuracy:*

 * Full Coverage: The ECAL, HCAL, and muon systems are designed to cover nearly the entire solid angle around the interaction point. This maximizes the chance of detecting all particles produced in the collision, minimizing the MET that is simply due to undetected particles within the detector acceptance.
 * Complementary Measurement: The ECAL, HCAL, and muon systems provide complementary measurements of different types of particles. This redundancy helps to improve the overall accuracy of the MET measurement and provides robustness against detector inefficiencies or failures.
 * Magnetic Field: The solenoid is essential for accurately measuring the momentum of charged particles. Without a magnetic field, you can only measure the energy of neutral particles in the calorimeters, and you would have no way to distinguish between a high-energy neutral particle and several low-energy charged particles.
 * Calibration and Resolution: Each sub-detector is carefully calibrated to ensure that its measurements are accurate. The energy and momentum resolution of each sub-detector is also crucial for minimizing the uncertainty in the MET measurement. Better resolution means more precise momentum measurements.

In summary, the combination of the ECAL, HCAL, and muon systems inside the solenoid magnet allows the CMS detector to measure the transverse momentum of nearly all detectable particles produced in a collision. The accuracy of the MET reconstruction depends critically on the performance of each of these sub-detectors and their ability to work together to provide a complete and accurate picture of the event. Any improvement in the resolution or acceptance of these sub-detectors directly translates to an improvement in the accuracy of the MET measurement.

TABLE AGENT

Table Data Preparation and the Table Agent

Agent Definition: The Table Agent is a hybrid retrieval-and-reasoning agent designed to answer scientific questions about tables in CMS SUS papers. It combines semantic and keyword-based search over table captions, headers, and enriched cell contents (with technical explanations) with structured JSON querying to retrieve both descriptive and quantitative information for the user's question.

Preparation Process

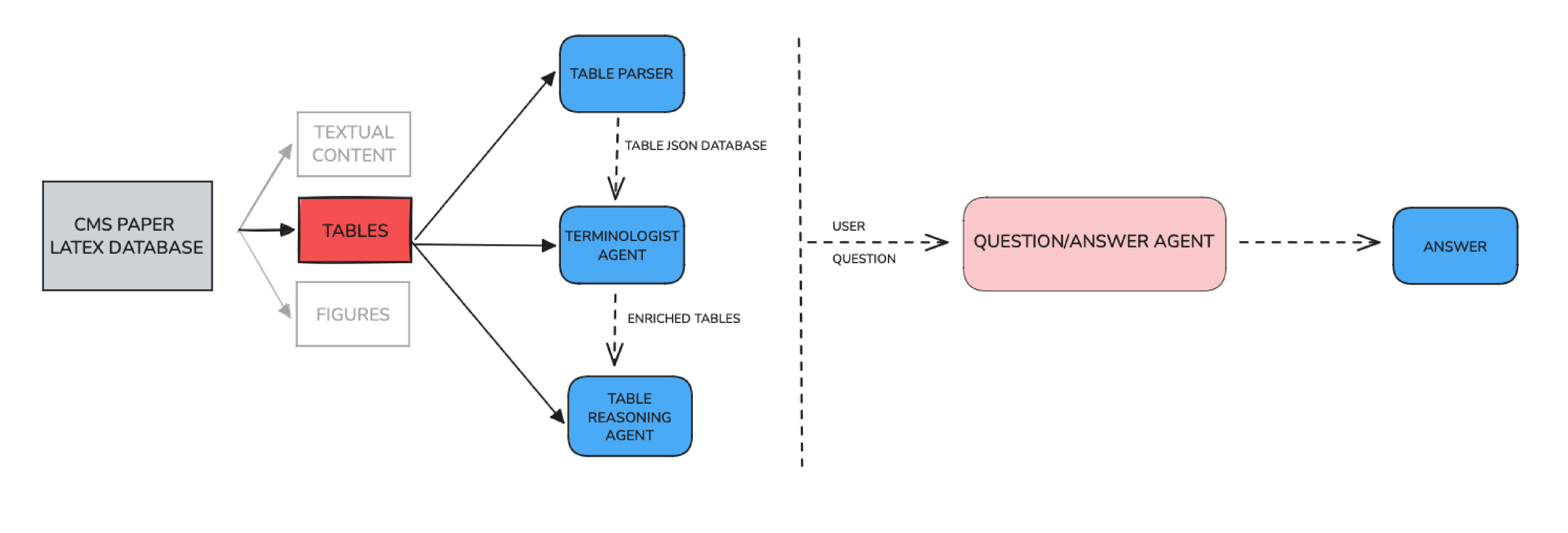


Table Database Preparation Process

Goal:
To structurally extract, enrich, and organize tables from CMS SUS papers so that both textual and quantitative queries can be answered efficiently.

Steps:

1. **Extract Tables:** All table blocks are parsed from the LaTeX source, with captions, headers, and rows converted into structured JSON format.
2. **Context Extraction:** For each table, the surrounding section titles and relevant paragraph contexts are captured and stored.
3. **Term Enrichment:** Every cell is scanned for LaTeX physics terms, which are converted into natural language explanations using the Terminologist Agent.
4. **Contextual Linking:** Table JSONs are enriched with section and paragraph context, linking each table to its scientific narrative and context.
5. **Store as Structured Data:** All enriched tables are saved into a searchable database (JSON, SQL, or vector store), enabling both text and structured queries.

Table Retrieval & Reasoning Pipeline

Goal:

To answer user questions about tables by combining structured retrieval and multi-step reasoning, re-using both tabular data and technical explanations.

Steps:

1. **User Question:** The system receives a scientific or technical question from the user.
2. **Retrieve Relevant Table(s) and Context:** Using semantic similarity and keyword matching over captions, headers, and enriched cell content, the agent retrieves the most relevant table(s) and their associated context.
3. **Table Q&A/Reasoning Agent:** The agent analyzes the user's intent—whether it is to find, compare, summarize, or infer—and determines which rows, columns, or context should be considered.
4. **Re-use Technical Explanations:** The agent integrates technical explanations (from the Terminologist Agent) into the answer, clarifying physics terms as needed.
5. **Step-by-Step Logic:** For calculations, comparisons, or inferences, the agent reasons step by step, making the logic chain explicit.
6. **Generate Comprehensive Answer:** The agent generates a well-structured, referenced answer, optionally citing the specific table, row, or term explanation.
7. **Provide Additional Guidance:** If the answer cannot be fully determined, the agent explains why and suggests what extra information might be needed.

Performance Test of Table Agent

Retrieval - 1

Question → If the integrated luminosity uncertainty increases to 5%, how would this affect the background estimation?

Answer →

ANSWER (step-by-step reasoning and technical explanation):

1. Relevant Tables and Terms:

Table 1 (Preselection criteria applied to all the events entering the SRs and CRs) is directly relevant.

The technical terms relevant are: ptmiss (missing transverse momentum), pt (transverse momentum), jets, multiplicity, eta (pseudorapidity), SRs (signal regions), and CRs (control regions).

2. Context and User's Goal:

The context is a search for dark matter using events with jets and missing transverse momentum (ptmiss). The analysis involves preselection criteria applied to events before they are considered for signal regions (SRs) or control regions (CRs).

The user wants to know if an event that satisfies all preselection criteria except the leading jet pt requirement can still be included in the analysis, and the reasoning behind this.

3. Table Examination:

Table 1 lists the preselection criteria.

Row 2 states: 'Jets: multiplicity >= 1, pt > 30 GeV, abs(eta) < 2.5'. This means that to pass the preselection, events must have at least one jet with transverse momentum (pt) greater than 30 GeV and absolute pseudorapidity (abs(eta)) less than 2.5.

Row 1 states: 'ptmiss (recoil in CR) > 250 GeV'. This requires the missing transverse momentum to be greater than 250 GeV.

4. Inference and Reasoning:

The preselection criteria in Table 1 act as a filter. Events must pass all criteria to be considered for further analysis in either the signal regions (SRs) or control regions (CRs).

If an event fails any of the preselection criteria, it is excluded from the analysis before it can be classified as signal or background.

The user's hypothetical event fails the leading jet pt requirement. This means that the event does not meet the condition 'pt > 30 GeV' for at least one jet.

5. Technical Term Re-use:

pt (transverse momentum) is a measure of a particle's momentum perpendicular to the beam axis. A higher pt indicates a more energetic particle moving transversely.

Jets are sprays of particles resulting from the hadronization of quarks or gluons produced in the collision.

ptmiss (missing transverse momentum) is the imbalance in momentum in the transverse plane, which can be an indicator of the presence of weakly interacting particles such as dark matter particles that escape detection.

6. Conclusion with Detailed Reasoning:

Before: All events enter an initial pool.

During: The preselection criteria are applied. The event in question fails the 'Jets: multiplicity >= 1, pt > 30 GeV, abs(eta) < 2.5' criterion because its leading jet has a pt less than or equal to 30 GeV.

After: Because the event fails to satisfy all of the preselection criteria defined in Table 1, it cannot be included in the SRs or CRs for further analysis. The preselection is designed to reduce background and select events with the desired characteristics (at least one energetic jet and significant missing transverse momentum), and failing any of these initial cuts means the event doesn't resemble the kind of event this analysis is designed to study.

7. Concise Summary:

An event failing the leading jet pt requirement in Table 1 cannot be included in the analysis, as events must satisfy all preselection criteria to be considered.

Performance Test of Table Agent

Retrieval - 2

Question → How would relaxing the cut on ptmiss from 250 GeV to 200 GeV likely affect the background and signal yields?

Output →

Below is a step-by-step, technical explanation in response to the user's question.

1. Relevant Tables and Terms

The main reference is the table titled 'Preselection criteria applied to all the events entering the SRs and CRs.' This table defines the selection cuts for events, including the ptmiss (missing transverse momentum) requirement, which is the focus of the user's question.

Key technical terms include:

- ptmiss (Missing transverse momentum): The magnitude of the vector sum of the momenta of all undetected particles in the transverse plane. High ptmiss is a classic signature of particles escaping detection, such as neutrinos or dark matter candidates.
- Signal Regions (SRs): Areas in the data phase space enriched in events from the targeted signal process (e.g., dark matter production).
- Control Regions (CRs): Areas enriched in background processes, used to estimate the background contribution in SRs.

2. Context and User Goal

This analysis searches for dark matter by selecting events with large missing transverse momentum (ptmiss) and one or two b-tagged jets. The user wishes to understand the impact of changing the ptmiss selection threshold from 250 GeV to 200 GeV, specifically whether this will increase or decrease the number of selected events and how the change will affect both signal and background.

3. Table Examination and Information Extraction

The preselection table specifies that events must have ptmiss > 250 GeV to proceed for further analysis in both signal and control regions. Lowering this cut to 200 GeV means events with ptmiss between 200 and 250 GeV, previously excluded, will now be included.

4. Inference and Reasoning

Background: Lowering the ptmiss threshold from 250 to 200 GeV will accept more events in the 200-250 GeV range. Since Standard Model processes (e.g., neutrinos from W/Z boson decays or mismeasured jets) can also create missing transverse momentum, the background yield will almost certainly increase. Control regions designed to estimate the background will also see higher event counts.

Signal: Signal regions are defined to select events from dark matter production, typically characterized by large ptmiss. Reducing the cut will increase the number of signal events, but the scale of this increase depends on the ptmiss spectrum of the signal. If the dark matter signal is mostly above 250 GeV, the increase may be small; if many signal events are in the 200-250 GeV range, the increase may be more substantial.

Signal-to-Background Ratio: Since background processes are generally far more frequent than signal, the increase in background is expected to outpace the increase in signal. As a result, the signal-to-background ratio will likely decrease, making it more challenging to identify a dark matter signal.

5. Clarification of Technical Terms

By reducing the ptmiss threshold, events with lower missing transverse momentum are included in both the signal and control regions, which can dilute the analysis's sensitivity to new physics.

6. Conclusion and Reasoning

Before the change, only events with ptmiss > 250 GeV were considered. After the change, events with ptmiss > 200 GeV are included. This change is likely to increase both background and signal event counts, but the background increase will likely be greater, thus decreasing the signal-to-background ratio and making a potential dark matter signal harder to distinguish from background.

7. Concise Summary

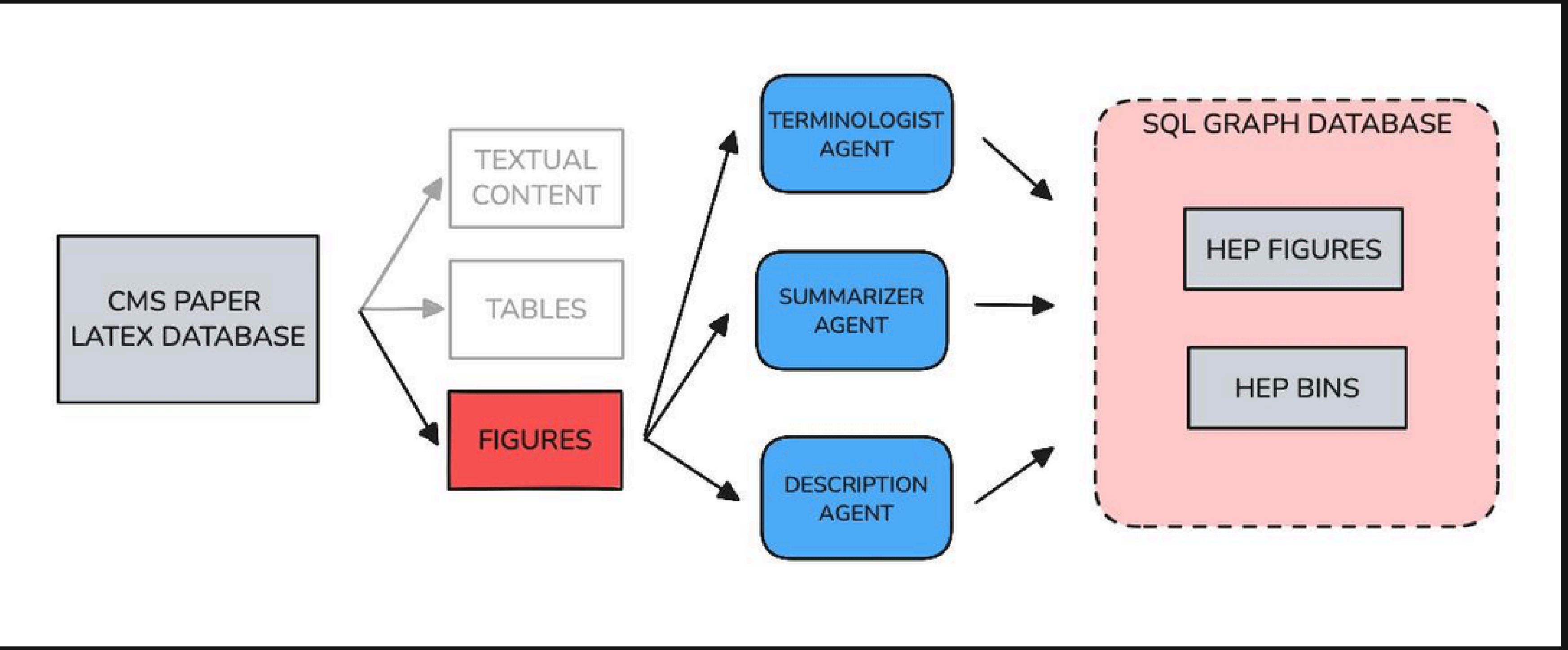
Relaxing the ptmiss cut from 250 GeV to 200 GeV will likely increase both background and signal yields, but the signal-to-background ratio will probably decrease, making a dark matter signal more difficult to detect.

GRAPH AGENT

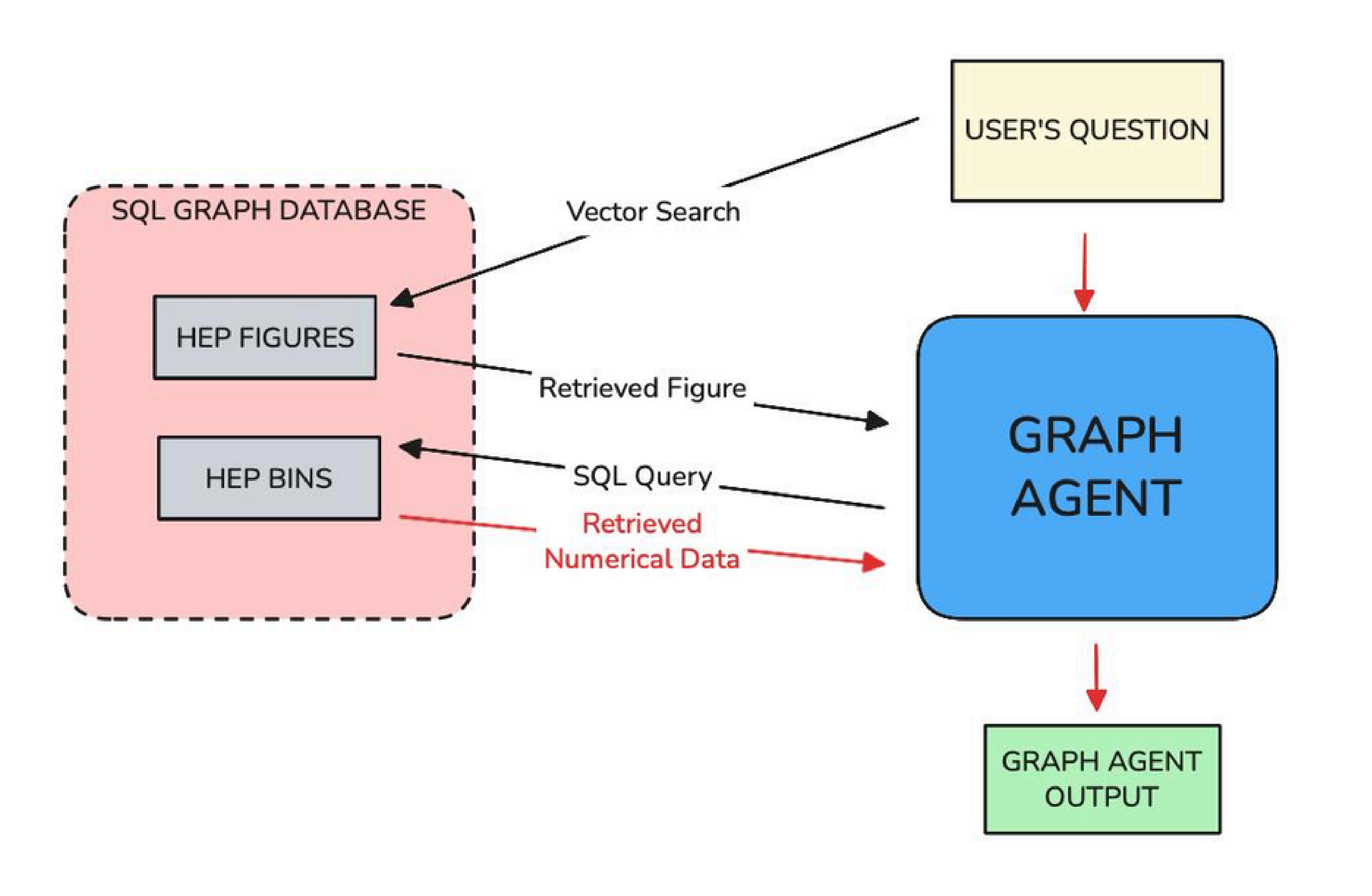
Graph Data Preparation and the Graph Agent

Agent Definition: The Graph Agent is a hybrid retrieval-and-reasoning agent designed to answer scientific questions about figures (graphs) in the CMS SUS papers. It combines semantic similarity search over figure metadata with SQL-based numerical querying to retrieve both descriptive and quantitative information from a structured database.

Graph Database Preperation Process



Retrieval Process of Graph Data with Grap Agent



Definition of Sub-Agents

Terminologist Agent → Converts LaTeX physics terms into natural language to ensure clean and interpretable figure metadata

Summarizer Agent → Generates concise summaries for each section and paper to provide contextual grounding for each figure

Description Agent → Produces optimized, search-friendly figure descriptions using axis info, metadata, and summaries

Graph Retrieval Pipeline

Step 1: Extract figures from the LaTeX source code of the CMS paper

Step 2: LaTeX physics terms in text and figure metadata are converted to natural language using the Terminologist Agent

Step 3: Each section and the full paper are summarized using the Summarizer Agent to provide contextual understanding for figures

Step 4: Figure JSONs are enriched with explanations and section summaries, linking plots to their scientific context

Step 5: Optimized, search-friendly descriptions for each figure are generated by the Optimized Description Agent using axis labels, summaries, and metadata

Step 6: Textual information is embedded using allenai-specter and stored in pgvector in the SQL database for semantic retrieval

Step 7: Figure metadata and numerical bin data are inserted into the SQL database

Step 8: The Graph Agent retrieves the most relevant figure via pgvector similarity search based on the user's question

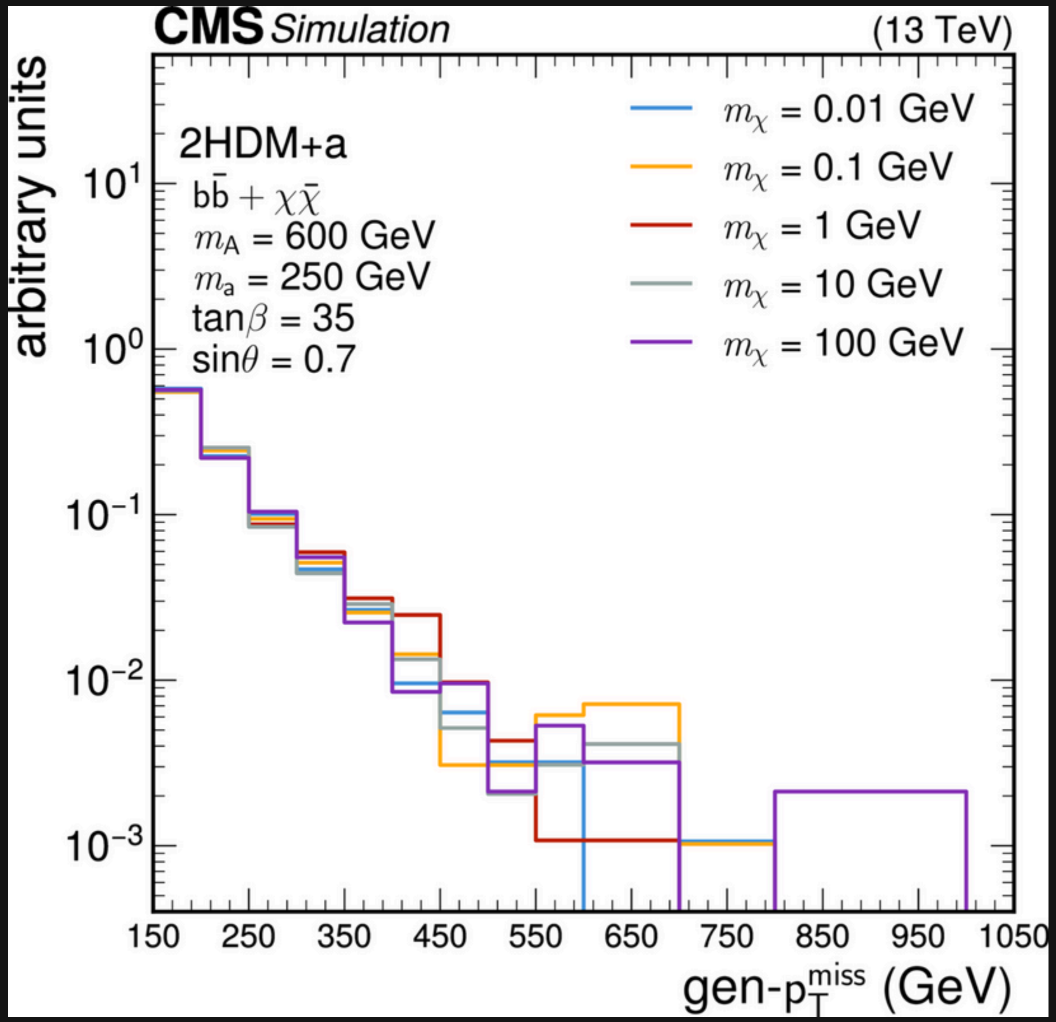
Step 9: If numerical data is needed, the Graph Agent generates an SQL query to extract values from hep_bins and combines all information to generate a natural-language answer using Gemini to be used in generation of the final answer

Performance Test of Graph Agent

Graph Retrieval

Question → Why do events with larger dark matter masses tend to produce more missing transverse momentum in this simulation?

Retrieved Graph:



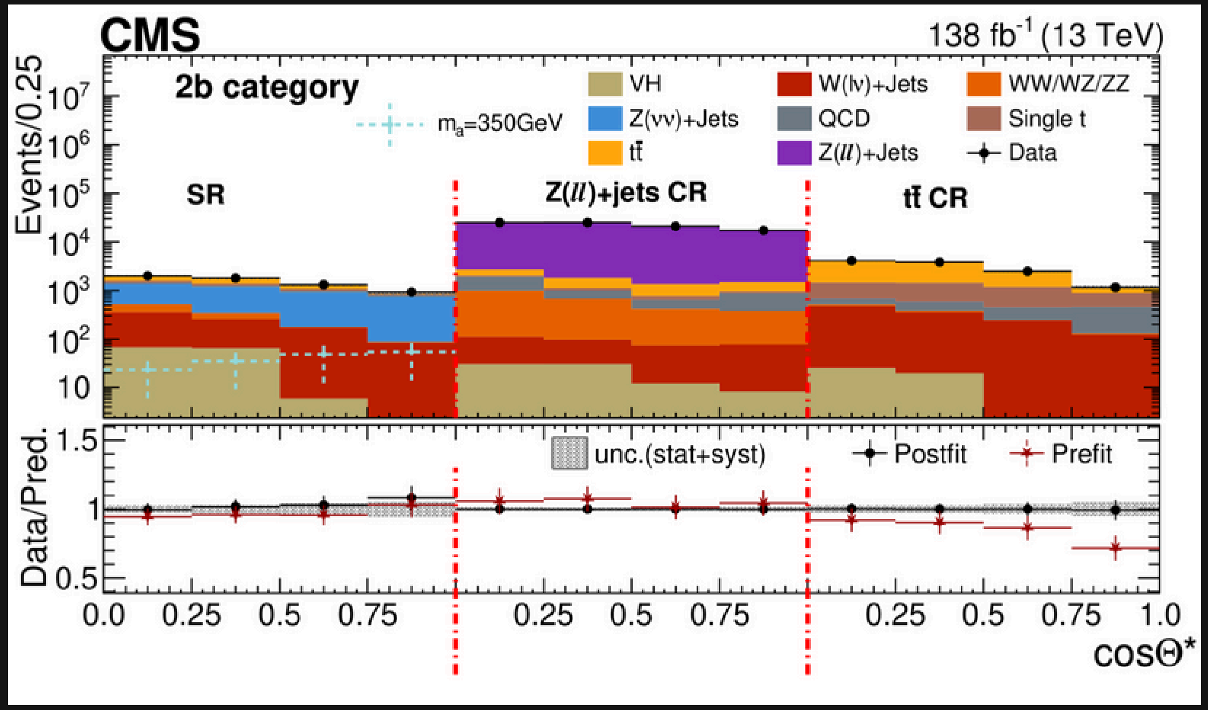
Numerical Value Retrieval

Question → In the Z to dilepton control region of Figure 6, what is the approximate number of observed events near cos(theta) = 0.5 in the Collins-Soper frame, and how closely does the postfit prediction agree with the data at that point?

SQL Query Generated:

```
SELECT x_bin, y_value, group_label FROM hep_bins WHERE figure_uid = 'cmssus23008_figure_6_z_to_dilepton_control_region'
AND CAST(x_bin->>'low' AS FLOAT) <= 0.5 AND CAST(x_bin->>'high' AS FLOAT) >= 0.5 AND group_label IN ('Data', 'Postfit')
```

Output→ In the Z to dilepton control region, near cos(theta) = 0.5 in the Collins-Soper frame, the number of observed events (Data) is approximately 20973. The postfit prediction at that point is 20968. Thus, the postfit prediction agrees very closely with the data.



Thank you...